

Julian Sanchez

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U.S. Top-Secret Security Clearance

Education

Florida State University - BSc, Computer Science

Expected May 2027

Seoul National University - Exchange Semester, Computer Science

Feb 2025 – Jun 2025

Experience

Software Engineer Intern – [Johns Hopkins Applied Physics Laboratory](#) – Laurel, MD

May 2026 – Present

- Working on internal C++/Python tools for the QKR Nuclear Communications Systems group, supporting NC3 initiatives.

Software Engineer Intern – [Pfizer](#) – New York, NY

Jun 2025 – Aug 2025

- Prototyped Python RAG tool with Ollama/Streamlit over a ChromaDB vector store for [ThisIsLivingWithCancer's](#) semantic search, restricting LLM responses to on-site articles and conversational summaries, improving user experience.
- Built Python red-team pipeline logging prompts and outputs to PostgreSQL, enabling automated safety benchmarking.
- Partnered with product managers through iterative development cycles to refine and clarify tooling requirements.

Software Engineer Intern – [MIT Lincoln Laboratory](#) – Lexington, MA

May 2024 – Aug 2024

- Implemented NASA System Assessment algorithm in Python, automating diagram-based readiness calculations at scale.
- Parsed Visio metadata using the vsdx library to auto-generate calculation matrices, eliminating manual configuration.
- Built a PyTest functional suite validating algorithm outputs against existing calculations, ensuring consistent results.

Software Engineer Intern – [Northrop Grumman](#) – Melbourne, FL

May 2023 – Aug 2023

- Extended internal Java/JavaFX avionics analysis tool to support additional flight-recording data types.
- Hardened data-processing reliability with Bamboo CI/CD regression testing, preventing downstream analysis failures.
- Participated in Agile Scrum development with sprint planning and daily standups across cross-functional teams.

Selected Projects

Rusteric – Rust

[GitHub](#)

- Created a linear algebra library on flat Vec, implementing core operations, decompositions (RREF, LU), and linear solvers.
- Implemented parallel matrix multiplication via Rayon, supporting multi-threaded computation of large matrix products.

Chess CLI – C++17

[GitHub](#)

- Implemented C++ chess game with polymorphic piece hierarchy, enabling rule-enforced move validation per piece type.
- Built a terminal chess interface with algebraic coordinate input and Unicode board rendering.

Starlink Observer – TypeScript, Next.js, Three.js, satellite.js, SpaceX API

[GitHub](#) | [Website](#)

- Developed a Starlink orbital visualization using satellite.js TLE propagation, rendered on a rotating Three.js globe.
- Implemented camera-follow tracking and orbital metadata panel, allowing per-satellite TLE element inspection.

Expense Insight – TypeScript, Next.js, Chart.js, PDF.js

[GitHub](#) | [Website](#)

- Built a client-side PDF parser with PDF.js, automating transaction extraction from Truist bank statements.
- Developed a multi-statement expense dashboard with Chart.js and Next.js, enabling categorical spending trend analysis.

Skills and Interests

- **Languages:** C++, Python, Java, Rust, TypeScript
- **Tools:** Git/GitHub, Docker, PostgreSQL, Supabase, Bamboo, Jupyter, Tableau
- **Frameworks/Libraries:** Streamlit, Ollama, PyTest, Tailwind CSS, NumPy, Pandas, Matplotlib
- **Interests:** Solo-travel, landscape photography (Sony a6400), kayaking, board games (Catan, Risk, Chess), cooking